

Multiple Choice (10 marks)

1. If the parabola $y_1 = x^2 + 2x + 7$ and the line $y_2 = 6x + b$ intersect at only one point, what is the value of b ?
 - (a) 7
 - (b) 3
 - (c) 12
 - (d) 4
2. Let $12-9+c=12$. What value of c makes the equation true?
 - (a) 0
 - (b) 3
 - (c) 9
 - (d) 12
3. What is a reasonable height for a doorway?
 - (a) 2 km
 - (b) 2 cm
 - (c) 2 m
 - (d) 2 mm
4. There are three real numbers x that are not in the domain of $f(x) = \frac{1}{1+\frac{1}{1+\frac{1}{x}}}$. What is the sum of those three numbers?
 - (a) 0.5
 - (b) 0
 - (c) -1
 - (d) -1.5
5. If $888x + 889y = 890$ and $891x + 892y = 893$, what is the value of $x - y$?
 - (a) 1
 - (b) -3
 - (c) -1
 - (d) 3

True / False (10 marks)

Answer True or False

6. True or False: The positive difference between the solutions of the quadratic equation is $5\sqrt{2}$.
7. True or False: 7% is equal to 0.07 when expressed as a decimal.
8. True or False: 26.1 mm is equal to 2.61 dm.
9. True or False: After 10 years, an investment of 4000 at the given growth rate is approximately 8,902.
10. True or False: The numbers 5 and 14 add up to 19 and multiply to give 70.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Let a, b, c be the roots of $x^3 + 2x^2 + 5x + 7 = 0$. Compute

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

12. Discuss the relationship between an object's weight on Earth and its weight on the moon. How can you express the weight on the moon in terms of its weight on Earth? Provide a detailed explanation.

End of Paper